

Assignment – React-Component-Program

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Question: A5: Create a simple React component that maintains a list of todos using useState. Allow the user to add a todo and display all added todos on the screen.

Source Code:

```
import React, { useState } from  
"react";
```

```
function TodoList() {
```

```
    const [todo, setTodo] =  
    useState("");
```

```
    const [todos, setTodos] =  
    useState([]);
```

```
    const addTodo = () => {  
  
        if (todo.trim() === "")  
        return;
```

```
        const newTodo = {  
  
            id: Date.now(),  
  
            text: todo.trim()  
  
        };
```

```
        setTodos([...todos,  
        newTodo]);
```

```

      setTodo("");
    };

    const deleteTodo = (id) => {

      const updatedTodos =
        todos.filter((item) => item.id
          !== id);

      setTodos(updatedTodos);

    };

    return (

      <div

        style={{

          width: "320px",

          margin: "20px auto",

          padding: "20px",

          border: "1px solid
#ccc",

          borderRadius: "8px"

        }}

      >

        <h2 style={{ textAlign:
"center" }}>Todo List</h2>

        <div style={{ display:
"flex", gap: "5px",
marginBottom: "15px" }}>

          <input

```

```

        type="text"

        value={todo}

        onChange={ (e) =>
setTodo(e.target.value) }

        onKeyDown={ (e) =>
e.key === "Enter" && addTodo() }

        placeholder="Enter a
todo"

        style={{ flex: 1,
padding: "5px" }}

    />

    <button
onClick={addTodo}>Add</button>

</div>

    <h3 style={{ textAlign:
"left", marginBottom: "10px" }}>

        Added Todos

    </h3>

    {todos.length === 0 ? (

        <p style={{ textAlign:
"left" }}>No todos added
yet.</p>

    ) : (

        <ul style={{ listStyle:
"none", padding: 0, margin: 0
}}>

            {todos.map((item) => (

                <li

                    key={item.id}

```

```

        style={{
            display: "flex",
            justifyContent:
"space-between",
            alignItems:
"center",
            padding: "6px
0",
            borderBottom:
"1px solid #eee"
        }}
    >
        <span>{item.text}<
/span>
        <button
onClick={ () =>
deleteTodo(item.id) }>
            Delete
        </button>
    </li>
    ) ) }
</ul>
    ) }
</div>

);
}

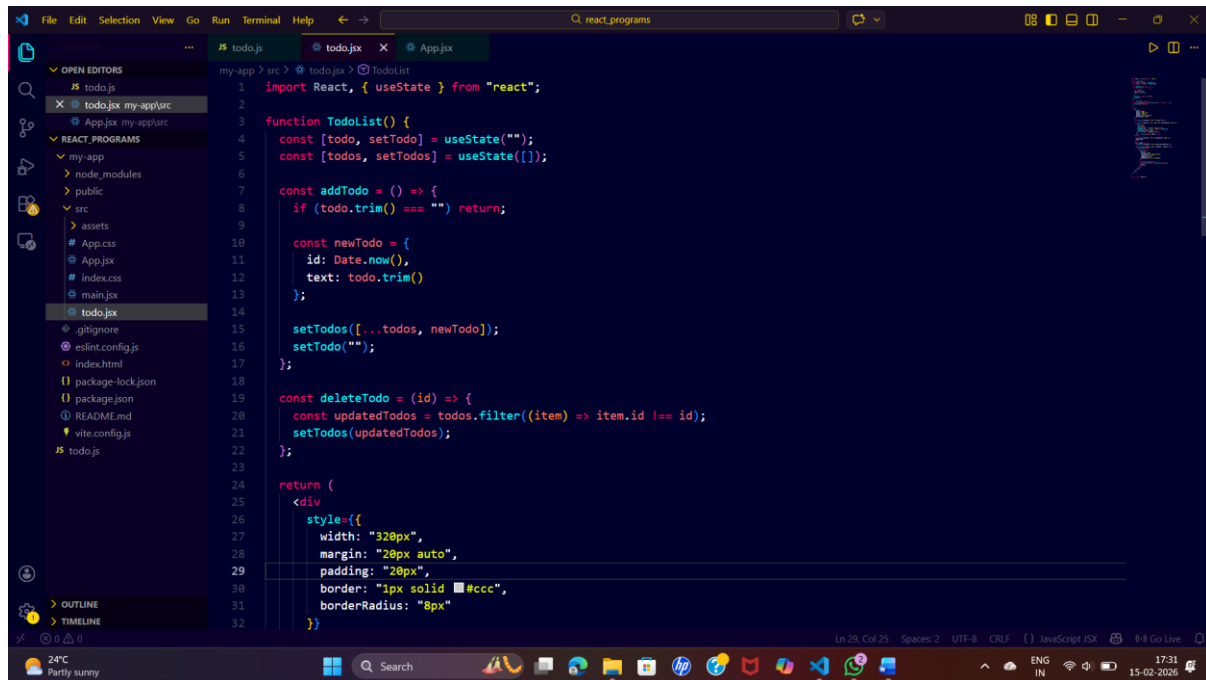
```

```

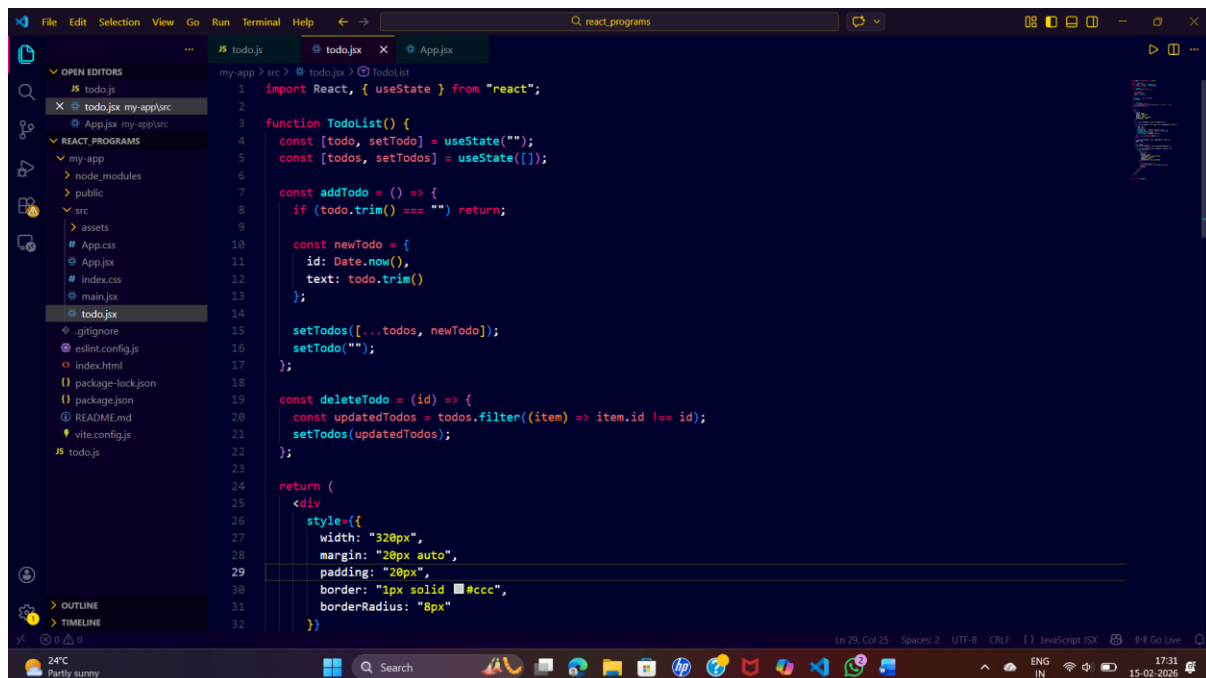
export default TodoList;

```

Source Code:



```
1 import React, { useState } from "react";
2
3 function TodoList() {
4   const [todo, setTodo] = useState("");
5   const [todos, setTodos] = useState([]);
6
7   const addTodo = () => {
8     if (todo.trim() === "") return;
9
10    const newTodo = {
11      id: Date.now(),
12      text: todo.trim()
13    };
14
15    setTodos([...todos, newTodo]);
16    setTodo("");
17  };
18
19  const deleteTodo = (id) => {
20    const updatedTodos = todos.filter((item) => item.id !== id);
21    setTodos(updatedTodos);
22  };
23
24  return (
25    <div
26      style={{
27        width: "320px",
28        margin: "20px auto",
29        padding: "20px",
30        border: "1px solid #ccc",
31        borderRadius: "8px"
32      }}
33    >
```



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```

The screenshot shows the VS Code editor with the file explorer on the left. The file explorer shows the project structure: my-app > src > App.jsx. The main editor displays the App.jsx file with the following code:

```
function TodoList() {  
  <h3>  
    Added Todos  
  </h3>  
  
  {todos.length === 0 ? (  
    <p style={{ textAlign: "left" }}>No todos added yet.</p>  
  ) : (  
    <ul style={{ listStyle: "none", padding: 0, margin: 0 }}>  
      {todos.map((item) => (  
        <li  
          key={item.id}  
          style={{  
            display: "flex",  
            justifyContent: "space-between",  
            alignItems: "center",  
            padding: "6px 0",  
            borderBottom: "1px solid #eee"  
          }}  
        >  
          <span>{item.text}</span>  
          <button onClick={() => deleteTodo(item.id)}>  
            Delete  
          </button>  
        </li>  
      )>  
    </ul>  
  )>  
}
```

The status bar at the bottom shows the file is at line 29, column 25, with 2 spaces, UTF-8 encoding, and CRLF line endings. The system tray at the bottom shows the date and time as 15-02-2026, 17:32.

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  const addTodo = () => {  
    if (todo.trim() === "") return;  
  
    const newTodo = {  
      id: Date.now(),  
      text: todo.trim()  
    };  
  
    setTodos([...todos, newTodo]);  
    setTodo("");  
  };  
}
```

The terminal at the bottom shows the following output:

```
PS C:\Users\gouri\OneDrive\Desktop\react_programs> cd my-app  
PS C:\Users\gouri\OneDrive\Desktop\react_programs\my-app> npm run dev  
  
> my-app@0.0.0 dev  
> vite  
  
Port 5173 is in use, trying another one...  
  
VITE v7.3.1 ready in 415 ms  
  
→ Local: http://localhost:5174/  
→ Network: use --host to expose  
→ press h + enter to show help
```

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Sample Output:

